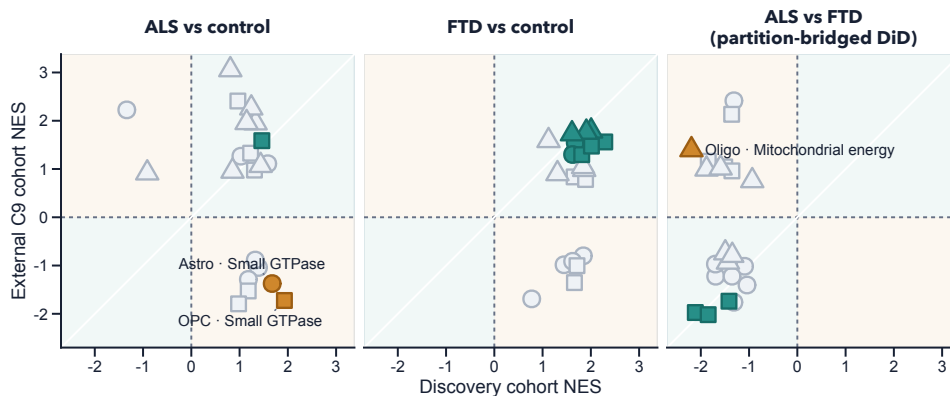


a Exact-estimand programme replication

All 63 matched axes: 39 share direction, 13 meet the strict dual-cohort gate, and 3 are significant and LOO-stable but discordant. Pale teal quadrants indicate direction agreement; pale amber quadrants indicate disagreement.

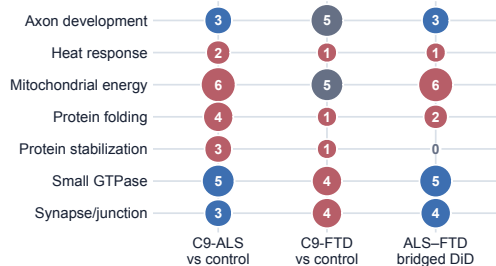


Evidence class Other matched axis Strict replicated Stable discordant

Cell type Astrocyte OPC Oligodendrocyte

b External C9 programme landscape

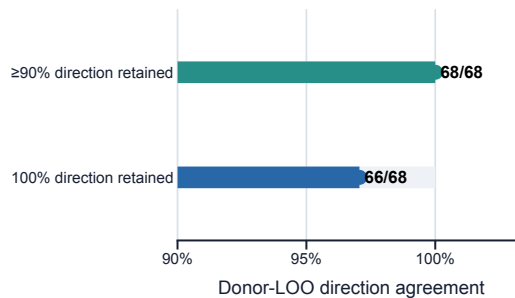
Global-FDR cell-type counts;
colour gives consensus direction.



Consensus direction positive negative mixed no significant axis

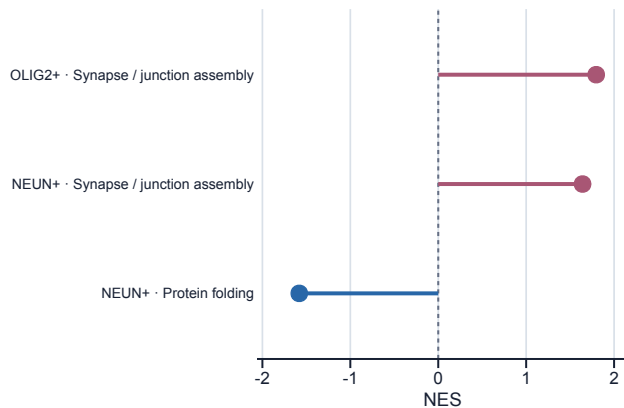
c Donor-LOO robustness

68 global-FDR axes;
65 had 37 valid refits, 3 had 36.



d GRN-FTD boundary

Three axes; each 9/9 LOO-stable. NES shown without CI.



e Candidate-gene boundary

Direction agreement across 9 settings per gene.
No axis passes the strict gene gate (0/153).

